

Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

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Abstract

Two failure modes can leave Conditional Value-at-Risk (CVaR)-constrained financial reinforcement learning (RL) unenforced without any visible symptom. The first is a **Lagrangian constraint-coupling failure**, wherein a scale mismatch between the reward and constraint channels (an unstandardised cost advantage, or a dual step mis-scaled to the CVaR excess) leaves the dual permanently inert, so that the CVaR budget is nominal rather than enforced. The second is an **evaluation-protocol reversal**: on a 31-asset universe, a single stress window ranks a learned allocator above every classical optimiser, while a 20-fold rolling walk-forward inverts that ranking. “Beat-the-optimiser” claims can therefore be protocol artefacts, although the constraint’s tail reduction survives both protocols (Wilcoxon $p = 6 \times 10^{-4}$). We name, diagnose, and resolve both failures on a CVaR-constrained portfolio allocator (seven-ETF macro universe, 2008–2024, weekly, 5 bps costs, 40-fold walk-forward, pooled Wilcoxon $p = 8.8 \times 10^{-6}$, or 3×10^{-4} with folds clustered by calendar, Benjamini–Hochberg (BH) and Deflated-Sharpe controlled). Scaling the dual correctly makes it bind in training, yet the budget can still be breached out of sample (71–78% across four validation-calibrated rules), which makes budget feasibility a second, separate knob. The remedy is a decision-time projection that estimates forward CVaR from trailing statistics and shrinks any non-compliant proposal toward a trailing minimum-variance anchor. At matched dual settings it reduces out-of-sample breach from 75.1% to 7.4%, matching the classical optimisers’ own 7.9% compliance rate, at Sharpe 0.85–0.99 across filtered arms against 0.90 for rolling minimum variance. Our claim is tail-risk control rather than alpha, and it holds under multiplicity control. The practical checklist is to put the reward and constraint channels on a common scale, verify that the multiplier moves off zero, treat budget feasibility as a separate knob, and enforce compliance at decision time.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → *Economics*.



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Keywords

portfolio allocation, reinforcement learning, Conditional Value-at-Risk, drawdown control, constrained MDP, differentiable optimisation

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1 Introduction

In the week of 16 March 2020, the S&P 500 fell another 15% over five sessions as COVID-19 lockdowns shut economies worldwide. A learned portfolio allocator trained on the preceding decade, with no explicit rule about how much loss it was budgeted to absorb, had no mechanical reason to rotate into Treasuries or cash when the VIX closed at a record 82.69. Absent a tail limit, the policy kept holding equities through the worst losses. An otherwise-identical allocator carrying an explicit Conditional Value-at-Risk (CVaR) limit had already rotated into the cash/rates leg before the trough and emerged with roughly half the drawdown (11.8% peak-to-trough versus 20.6% for its unconstrained twin, Table 3).

This difference is a result, although not the main finding of the paper. An explicit CVaR constraint is a safety layer that cuts the tail losses of a *learned* allocator without sacrificing risk-adjusted return. In the first model-free implementation we tried, a standard Lagrangian actor–critic, the *dual channel was inert*: the multiplier stayed near zero, so the mechanism meant to enforce the constraint exerted no tail pressure. Section 5 traces this to a scale mismatch likely present in any Lagrangian RL system that standardises the return advantage but not the cost advantage. The constrained *pipeline* still regularises the tail through CVaR-feasible checkpoint selection and the admissible-set projection, even when the multiplier is inert.

Reinforcement learning for portfolio allocation is usually evaluated on return or Sharpe ratio, with risk handled implicitly. Practitioners, however, manage *tail* risk explicitly. Markowitz mean-variance optimisation [19] penalises upside and downside symmetrically and is silent on path-wise drawdown. CVaR is a coherent, convex tail measure [21] that RL can optimise directly. We therefore ask a narrow, falsifiable question.

Does an explicit CVaR constraint reduce the downside tail losses and constraint breaches of a learned allocator, versus an otherwise identical unconstrained learner,

while remaining competitive with standard portfolio optimisers after costs?

We make the following contributions.

- (1) **Two named failure modes in CVaR-constrained financial RL, with diagnostics and, where demonstrated, fixes.** A *Lagrangian constraint-coupling failure*: a scale mismatch between the reward and constraint channels leaves the multiplier inert, through an unstandardised cost advantage in the model-free actor-critic and through a dual step mis-scaled to the CVaR excess in the differentiable allocator. The mismatch is specific to portfolio allocation, where weekly returns are $O(10^{-3})$ and the constraint signal is $O(10^{-2})$ to $O(1)$ (Section 5). An *evaluation-protocol reversal*: on a 31-asset universe a single stress window ranks the learned allocator above every optimiser, while a rolling walk-forward inverts that ranking, so “beat-the-optimiser” results can be protocol artefacts (Section 7.8).
- (2) **A differentiable CVaR-constrained allocator whose tail effect survives multiplicity control.** On a leak-free, tested portfolio-control environment, a return objective plus a differentiable CVaR penalty is back-propagated through the rollout, with a Lagrangian dual on a breach-rate budget and a residual policy over an adaptive anchor (Section 4). The tail-risk reduction holds on a 40-fold, two-universe walk-forward under Benjamini-Hochberg and Deflated-Sharpe control (Section 7.4), and a regime-switching hybrid converts it into a walk-forward Sharpe gain of 1.316 vs 1.209 for pure min-variance, with overlapping bootstrap intervals (Sections 7, 8).
- (3) **An evaluation protocol built to expose exactly this kind of artefact.** Deterministic optimisers plus three tuned model-free RL families are compared under a two-universe rolling walk-forward with Wilcoxon, Benjamini-Hochberg, and Deflated-Sharpe control. This design is what surfaces the protocol reversal in Contribution 1. The regime-switching hybrid is a supporting observation within this protocol rather than a separate contribution.

The pure learned allocator does **not** beat a rolling minimum-variance optimiser in aggregate, and we do not claim that it should. The *constrained* allocator’s value lies in tail control, which the regime-switching hybrid converts into an out-of-sample Sharpe gain (+0.107) in the high-volatility regime where the constraint is most active. The broader contribution is a differentiable constraint mechanism that demonstrably binds and cuts tails, together with two transferable failure modes that any study of constrained RL in finance must control for, with the allocator as the testbed for both.

2 Background and related work

Mean-variance optimisation and its inputs. Mean-variance optimisation (MVO) [19] is sensitive to input estimates. Decision-focused learning trains return predictors to improve downstream MVO decisions rather than raw forecast accuracy [17], and mean-variance-efficient collaborative filtering blends user preference with portfolio theory [9]. Both improve the *inputs* to a variance-based objective. We instead constrain a *tail* risk measure on a learned policy.

Conditional Value-at-Risk. CVaR_α is the expected portfolio loss on the worst $(1-\alpha)$ fraction of outcomes, the average of losses exceeding the α -quantile (VaR_α). Unlike variance, it ignores the upside, and unlike VaR, it averages across the whole tail. Rockafellar and Uryasev [21] give the tractable form

$$\text{CVaR}_\alpha(L) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[\max(L - \zeta, 0)] \right\}, \quad (1)$$

where L denotes the portfolio loss and the inner expectation is over the empirical loss distribution. The minimum is attained at $\zeta = \text{VaR}_\alpha$, and the representation admits both a convex linear program (LP), included as a baseline, and a differentiable penalty via topk .

RL for portfolio management. The dominant pattern in the RL-for-portfolio literature is to train on a single historical window, evaluate on one held-out test split, and compare against equal-weight or buy-and-hold. EIIE [15] compares twelve classical strategies on one ≈ 50 -day crypto window, and SARL [25] reports higher return on a fixed period without cross-validation. FinRL [18] supports rolling retraining but demonstrates it on a single split with no significance tests. Huang [14] applies continuous-time RL to asset-liability management, although with a quadratic surplus objective rather than a tail constraint. None of these uses CVaR as a constraint, seeds multiple runs, or applies statistical testing. Empirical finance has long documented how the selection of backtests inflates reported performance [3, 13, 24], and our protocol carries those controls over to RL portfolio allocation. We are not aware of prior work that subjects RL portfolio allocation to a 40-fold, two-universe walk-forward with Wilcoxon and Benjamini-Hochberg control, which is the protocol that exposes the reversal artefact.

Direct reinforcement and pathwise gradients. Moody and Safell [20] first trained a recurrent trading policy by back-propagating the *differential Sharpe ratio* directly through the return function. Buehler et al. [7] (Deep Hedging) extend this to neural derivative hedging, back-propagating CVaR as a risk-aversion *utility* through a simulated trajectory. Our allocator inherits the pathwise-gradient insight from both but replaces the risk-aversion coefficient with an explicit CVaR *budget constraint* enforced by a dual variable. Model-free policy gradients (Advantage Actor-Critic, A2C, and Proximal Policy Optimization, PPO [22]) and off-policy actor-critics (Soft Actor-Critic, SAC [12]) ignore this differentiable structure. Our allocator exploits it, which matters at this sample size (887 weekly observations, Section 7.7). ProtoHedge [11] similarly makes such policies auditable via a prototype-similarity policy.

CVaR-constrained RL. Chow et al. [8] derive the Lagrangian gradient for CVaR-constrained CMDPs. Tessler et al. [23] (RCPO) and Ying et al. [26] (CPPO) implement hard CVaR constraints for robotics control tasks. None identifies the *constraint-coupling failure* that we describe, wherein standardising the return advantage to unit scale without normalising the cost advantage leaves the Lagrange multiplier permanently inert. The failure is specific to portfolio allocation, where returns are $O(10^{-3})$ and breach indicators are $O(1)$, and it would remain invisible in robotics. We pose allocation as a constrained MDP [1, 2] and show that a well-scaled dual is what makes the constraint bind. The closest concurrent work is Dong [10] (CLAUSELENS), a CVaR-constrained RL system for reinsurance

pricing. Our contribution is orthogonal to it, in that we name failure modes that transfer to any CVaR-constrained financial RL system.

3 Problem formulation

Markov Decision Process (MDP). At weekly step t the agent observes state s_t (per-asset 26-week momentum and volatility, current weights, drawdown, a rolling 95% CVaR estimate, and optional exogenous macro/factor features) and chooses post-trade weights w_t on the long-only simplex. The environment projects w_t onto the admissible set $\mathcal{W}$ (max weight 0.40, turnover cap 0.50, optional cash floor), charges proportional cost $c\|w_t - w_{t-1}\|_1$ ($c = 5$ bps), and realises $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$, where ρ_t are next-week asset returns. Features at t use only $\rho_{<t}$ (no look-ahead, enforced by tests).

Constrained objective (CMDP). A *constrained Markov Decision Process* (CMDP) [2] augments the standard MDP with one or more inequality constraints that the policy must satisfy on average. We maximise risk-adjusted return subject to a tail budget, $\max_\pi J(\pi)$ s.t. $\text{CVaR}_\alpha(\text{loss}) \leq d$ with $\alpha = 0.95$. The Lagrangian relaxation converts the constraint into a penalty, $\mathcal{L}(\pi, \lambda) = J(\pi) - \lambda (\text{CVaR}_\alpha - d)$, and a dual variable $\lambda \geq 0$ is raised whenever the constraint is violated and lowered when slack remains. In equilibrium, λ is the shadow price of tail risk: a higher λ means that the policy has been pushing against its tail budget.

4 Method

Differentiable allocator. The actor is an MLP mapping s_t to simplex weights via softmax. Because the environment’s reward and a sample-CVaR are differentiable in w , we train by back-propagating, over randomised windows of the training series, the objective

$$\mathcal{L} = -\text{Sharpe}(\text{net returns}) + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad (2)$$

where $\text{Sharpe} = \bar{r}/\sigma_r$ is the annualised mean return divided by its standard deviation (reward per unit of risk), $\widehat{\text{CVaR}}$ is the mean tail loss over the worst $(1-\alpha)$ fraction of window net returns (a differentiable tail mean via topk), and d is the pre-specified tail budget. For this problem, the pathwise gradient is far more sample-efficient than score-function RL.

Pathwise gradient and where the constraint acts. Every term in $\mathcal{L}$ is a differentiable function of the rollout, so the policy is trained by exact back-propagation through the simplex map and the reward, without a score-function estimator. With net returns $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$ and tail set $\mathcal{T} = \{t : R_t \text{ among the worst } \lceil (1-\alpha)T \rceil \text{ of the window}\}$,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_t \left(\underbrace{-\frac{\partial \text{Sharpe}}{\partial r_t}}_{\text{all weeks}} - \underbrace{\frac{\lambda}{|\mathcal{T}|} \mathbf{1}[\widehat{\text{CVaR}} > d] \mathbf{1}[t \in \mathcal{T}]}_{\text{tail weeks only}} \right) \frac{\partial r_t}{\partial w_t} \frac{\partial w_t}{\partial \theta}, \quad (3)$$

where $\partial w_t / \partial \theta$ flows through the residual softmax (below). The topk subgradient routes the dual pressure to *exactly* the worst- $(1-\alpha)$ weeks, and only while the constraint is breached. The penalty lifts the deep-loss weeks without distorting the bulk of the path. This locality is what makes the constraint bind cleanly once the coupling is correctly scaled (Section 5).

Table 1: Hybrid vs. min-variance Sharpe gain, threshold sweep (15 configs, no retraining). Bold marks configurations where the hybrid is ahead. The default (0.75, −0.05) is the reported configuration.

selloff \ vol. quantile	0.65	0.70	0.75	0.80	0.85
−0.04	+0.202	+0.230	+0.161	+0.225	+0.058
−0.05	+0.154	+0.179	+0.107	+0.169	+0.002
−0.06	−0.007	−0.017	−0.086	−0.034	−0.183

Residual policy over an adaptive anchor. To generalise across regimes rather than overfit one price path, weights are $w_t = \text{softmax}(\log a_t + f_\theta(s_t))$, where a_t is an adaptive inverse-volatility anchor computed from recent returns. At initialisation $f_\theta \approx 0$, so the policy is the anchor (a strong, regime-robust prior), and training then learns a state-dependent tilt. Model selection keeps the checkpoint with the best validation objective among those that satisfy the validation CVaR limit, and weight decay is applied, both to curb backtest overfitting.

Baselines. Deterministic, re-estimated weekly: cash, equal weight, inverse vol, minimum variance, mean-variance, risk parity, and a min-CVaR Rockafellar–Uryasev LP [21]. The covariance-based rules use the sample covariance of the trailing 104 weeks and inverse volatility the trailing 52 weeks, all long-only under the learner’s weight cap. Learned: model-free A2C, PPO [22] (clipped surrogate, generalised advantage estimation, GAE), SAC [12] (off-policy, twin critics, auto-tuned entropy), and the unconstrained differentiable allocator ($\lambda = 0$).

Regime-switching hybrid. We construct a no-retrain hybrid: label each out-of-sample (OOS) week *high-volatility* if trailing 13-week SPY volatility meets the 75th-percentile cutoff (15% of OOS weeks), *selloff* if cumulative return $\leq -5\%$, else *calm*. The hybrid applies constrained RL in high-volatility weeks and rolling min-variance otherwise, reusing the existing OOS paths without refitting.

Threshold sensitivity. Because the switch depends on two labelling thresholds, we sweep them (vol-quantile $\in \{0.65, \dots, 0.85\}$, selloff $\in \{-0.04, -0.05, -0.06\}$, 15 configurations, no retraining, Table 1). The hybrid beats pure min-variance in **10/15** configurations (mean gain +0.077). It wins at every volatility cutoff when the selloff threshold is −0.04 or −0.05, but loses at all five cutoffs once the selloff threshold deepens to −0.06, because that threshold reclassifies the high-volatility weeks in which the constrained allocator adds most as selloffs and routes them to min-variance instead. The reported +0.107 at the default (0.75, −0.05) is representative rather than selected post hoc: the edge holds for volatility-defined stress but not once the selloff label absorbs it.

5 Decomposing the constraint: dual, architecture, and budget feasibility

We designed a three-arm diagnostic on the differentiable allocator to isolate what a Lagrangian CVaR constraint contributes, and where it can fail silently. All three arms share the identical pipeline (residual actor, CVaR penalty with dual ascent, CVaR-feasible checkpoint selection on validation, admissible-set projection) and training budget (5 seeds, 1500 updates). Only the dual’s learning rate η_λ and the presence of the constraint differ (Table 2). The

mis-scaled dual ($\eta_\lambda = 0.001$) takes steps far smaller than the CVaR excess it tracks requires. The **scaled dual** ($\eta_\lambda = 5.0$) matches the step to that scale. The **unconstrained** arm removes the CVaR penalty and selects on validation return alone.

Finding 1: dual inertness from a scale mismatch. With the mis-scaled dual, the multiplier stays near zero throughout training (mean final $\lambda = 0.022$ vs. 0.101 for the scaled dual). The dual ascends on the CVaR excess, $\lambda \leftarrow \max\{0, \lambda + \eta_\lambda(\widehat{\text{CVaR}} - d)\}$, which is $O(10^{-2})$ in weekly-return units, so a step of 10^{-3} cannot move λ within the training budget. The model-free actor-critic shows the same failure through its advantages. With the reward advantage standardised and the cost advantage left at its raw $O(10^{-3})$ scale, the constrained and unconstrained agents trained identically until we standardised the cost advantage, after which λ rose to $O(1)$. The diagnostic transfers directly: multiplier telemetry that stays flat at zero throughout training means that the constraint is decorative, whatever the loss curve shows.

Finding 2: most of the tail control is architectural, not the dual. With λ pinned near zero, the constrained pipeline alone (CVaR-feasible checkpoint selection over the residual anchor, together with the projection) still cuts CVaR₉₉ by 61% relative to the unconstrained learner (0.025 vs 0.065, 5-seed mean) and raises Sharpe by 58% (0.987 vs 0.626). A permutation test on the 5-seed distributions confirms the difference (observed -0.039 , one-sided $p = 0.0044$, two-sided $p = 0.0083$, $n = 50,000$). The scaled-dual arm produces a similar reduction relative to the unconstrained learner ($p = 0.0037$), so the gap does not depend on the dual engaging. Engaging the dual correctly ($\lambda = O(0.1)$) leaves CVaR₉₉ essentially unchanged versus the near-inert mis-scaled dual (0.028 vs 0.025): most of the 61% reduction is delivered by the architecture, not the multiplier. This is a positive result, since a constrained pipeline regularises the tail on its own, independently of the Lagrangian penalty. It is also what makes the coupling failure silent: tail metrics improve with or without a working dual, so only the multiplier telemetry reveals that the dual is not doing the work.

Finding 3: budget feasibility is a second knob, and it needs enforcement, not another calibration. The scaled dual binds in training, but at the tight budget ($d = 1.2\%$) produces a *higher* OOS breach rate (75%) than the mis-scaled dual (8%), though classical rules clear the same budget in 92% of weeks (Section 5). The scaled dual is in fact worse than the mis-scaled dual on every metric in Table 2, and model selection explains why. Scaling the dual lets most checkpoints meet the validation budget (27 of 30 on average, against 8 of 30), so selection keeps a later and more return-seeking checkpoint (update 1010 on average, against 370), and that checkpoint breaches under the test window’s tail shift. Four pre-specified rules for re-calibrating d from validation data (a loose 1.92% budget, matching the policy’s own or the anchor’s validation-realised CVaR_{0.95}, and a per-fold adaptive budget over a 10-fold walk-forward) all land in a 71–78% OOS breach band: a return-seeking Lagrangian gives up compliance for return under tail shift whatever d is, because the budget itself was never the source of the failure.

The fix: enforce the budget rather than penalise its violation. We therefore enforce the budget with a decision-time CVaR *filter*. Between the actor’s proposal and the environment, we estimate forward CVaR from a trailing covariance and mean only, both computed from returns strictly before t ($\text{CVaR} \approx c_\alpha \sigma_p - \mu_p$, where μ_p and σ_p are the trailing portfolio mean and volatility and $c_\alpha = \varphi(\Phi^{-1}(\alpha))/(1 - \alpha)$ is the Gaussian CVaR multiplier, $c_{0.95} = 2.06$). If the estimate exceeds d we shrink the proposal along the line toward a trailing minimum-variance anchor, solving numerically for the largest mixing weight that clears the budget, then trade the shrunk weights, with turnover/max-weight/long-only composed after as before. The decision at t depends only on the information set $\mathcal{F}_{t-1}$: the trailing mean, covariance and minimum-variance anchor use the $L = 104$ returns $\rho_{t-L}, \dots, \rho_{t-1}$, and the actor’s proposal uses features built from $\rho_{<t}$. A look-ahead audit (tests/test_risk_filter_no_lookahead.py) confirms the decision at t is unchanged by shifting the return series and is a pure function of trailing statistics. Both checks pass with zero deviation. We train with the filter active.

Four arms mirror the coupling arms exactly (5 seeds, 1500 updates, identical splits), changing only the named quantities (Table 2, bottom block, Figure 1). **A6, filter + scaled dual** (our intended method) cuts breach from 75.1% to 7.4% against the scaled dual at the same η_λ , going *below* every classical rule, with CVaR₉₉ 0.016 and mean excess 0.0043, at Sharpe 0.851. **A5, filter only** ($\eta_\lambda = 0.001$) coincides with the mis-scaled row. That policy already sits within budget, so the filter never triggers (0 of 1285 test weeks across seeds). It reaches the classical 7.9% at Sharpe 0.987 against minimum variance’s 0.901 and breaches less than half as deeply as minimum variance (mean excess 0.0126 vs. 0.0294). **A7, adaptive dual** (λ updated online from realised past losses, filter off) is the negative control: breach stays at 81.9% even as λ climbs to a mean of 7.69, so test-time dual adaptation alone enforces nothing, and the compliance gain comes from the filter rather than from the adaptation. **A8** recovers most of the compliance (14.6%), although less reliably, ranging from 7 to 33% across seeds against zero seed-to-seed variance for A5 and A6.

The filtered arms sit in the low-volatility region. A5 has a mean weight-space L_1 distance of 0.476 to rolling minimum variance, a correlation of 0.786, and a different top holding in 9.4% of rebalances, against 0.890, 0.201 and 97.5% for the unfiltered scaled dual. The filtered policy is therefore *min-variance-adjacent* without being a redundant way to compute minimum variance. From that region it dominates minimum variance on both axes, with higher Sharpe at an equal breach rate and half the breach depth (A5), and a lower breach rate outright (A6). A soft Lagrangian cannot enforce a tail budget out of sample, whereas a decision-time projection can.

Is the budget infeasible, or is the policy failing to reach it? When every classical baseline is re-scored against the identical rolling-CVaR_{0.95} budget on the same realised path, the breach rate is 7.9% for minimum variance, inverse volatility and the min-CVaR LP, 62.8% for equal weight and 75.3% for risk parity, against 75.1% for the scaled-dual RL. The budget is therefore reachable. The soft-penalty learner was not reaching it, and the filter arms now match the classical rate from a learned policy.

Table 2: Coupling ablation and decision-time filter, stress window (train pre-2018, test 2020/2022), 5-seed means. Breach rate: share of test weeks whose rolling 52-week $\text{CVaR}_{0.95}$ exceeds d (classical floor 7.9%, §5). Scaling the dual binds λ in training but does not fix OOS feasibility. The filter cuts breach from 75.1% to 7.4% at the scaled dual (A6). A5 equals the mis-scaled row because its filter never triggers. [†]Mean online-updated λ at evaluation.

variant	Sharpe	CVaR ₉₉	breach rate	final λ
unconstrained (baseline)	0.626	0.065	1.000	0.000
mis-scaled dual ($\eta_\lambda = 0.001$)	0.987	0.025	0.079	0.022
scaled dual ($\eta_\lambda = 5.0$, $d=1.2\%$)	0.911	0.028	0.751	0.101
scaled dual, loose budget ($d=1.92\%$)	0.543	0.050	0.780	0.046
A5 filter only ($\eta_\lambda = 0.001$)	0.987	0.025	0.079	0.003
A6 filter + scaled dual ($\eta_\lambda = 5.0$)	0.851	0.016	0.074	0.135
A7 adaptive dual ($\eta_\lambda = 5.0$ online)	0.826	0.031	0.819	7.692 [†]
A8 filter + adaptive dual	0.800	0.020	0.146	0.657 [†]

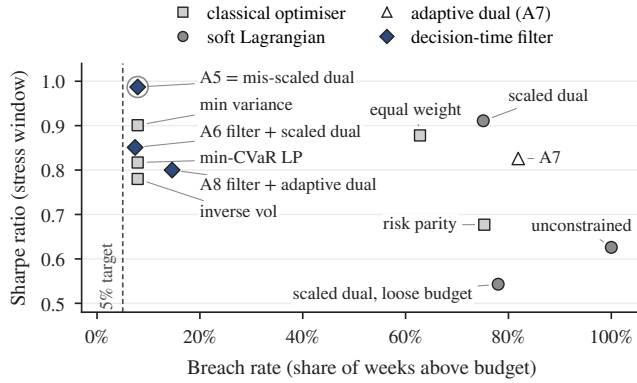


Figure 1: Compliance frontier, stress window. The dashed line is the 5% target. Classical optimisers (squares) and filter arms (diamonds) achieve the nearest observed breach rates (7.4–7.9%) at high Sharpe, while soft Lagrangians (circles) and the filter-off adaptive dual (triangle) remain far to the right. A5 overlaps the mis-scaled dual (ringed): its filter never fires.

6 Data

The primary universe consists of seven liquid exchange-traded funds (ETFs) that span the major macro blocks: SPY (equity), TLT (rates), HYG (credit), DBC (commodity), GLD (gold), UUP (US dollar) and BIL (cash). Adjusted closes from Yahoo are resampled to **887 weekly returns over 2008–2024**. Splits are chronological (60/20/20): train ≈ 2008 –2018 (incl. the 2008 global financial crisis, GFC), validation ≈ 2018 –2021, test ≈ 2021 –2024. For the stress study we retrain on 2008–2018 and test on the 267-week window containing the 2020 COVID crash and the 2022 selloff. Macro covariates (10y–3m term spread, VIX) are sourced from index proxies ($\sim \text{TNX}$ – $\sim \text{IRX}$, $\sim \text{VIX}$). A Federal Reserve Economic Data (FRED) loader is also provided. Portfolios are long-only with a maximum weight of 0.40, a turnover cap of 0.50 and gross exposure 1.0. Training uses 150 episodes and seeds {7, 13, 23, 42, 2025}, and significance uses a 2000-sample block bootstrap (block 4), Wilcoxon, BH and Deflated-Sharpe.

For the cross-universe significance test (Section 7.4) we add a second universe of ten S&P sector ETFs (XLK, XLF, XLE, XLV, XLY,

Table 3: Stress window (train pre-2018, test through 2020 & 2022), mean $\pm$ std over 5 seeds. Best in bold. Breach rate: share of weeks whose rolling 52-week $\text{CVaR}_{0.95}$ exceeds the environment’s 3% limit (Table 2 scores the 1.2% training budget). Violations: weeks per run breaching a portfolio constraint.

metric	unconstr.	constr.	change
Sharpe	0.63 $\pm$ 0.09	0.88$\pm$0.04	+40%
Sortino	0.61	0.87	+44%
Calmar	0.32	0.48	+50%
Max drawdown	0.206	0.118	–43%
CVaR ₉₅	0.0333	0.0189	–43%
CVaR ₉₉	0.0647	0.0327	–49%
Breach rate	0.234	0.189	–19%
Violations	7.2	0.0	–100%

XLP, XLU, XLI, XLB, plus a BIL cash leg), identical in every other respect (same provider, clock, window and cost). The two universes therefore share a calendar, and Section 7.4 clusters their folds accordingly. For the evaluation-protocol experiment (Section 7.8) we add a **third universe of 31 cross-asset, sector, and industry ETFs** (4 broad equity, 9 S&P sectors, 7 industry, 6 fixed-income, 3 commodity, FX, cash) with the same provenance, clock and cost. Its higher dimensionality stresses covariance estimation.

7 Results

Sharpe ratios for the constrained allocator (0.87 to 0.99) differ across tables because configurations differ: 1,500 updates over 5 seeds for the coupling and filter arms, 2,500 over 5 seeds for the stress study, 2,000 over 3 seeds for the ablations, and a 3-seed ensemble per walk-forward fold.

7.1 Calm chronological split: the constraint does not bind

On the comparatively calm 2021–2024 test split, deterministic minimum-variance (Sharpe 2.52, CVaR_{95} 0.0064) and the min-CVaR LP (2.24) dominate. The differentiable allocator reaches Sharpe 1.24 (vs equal weight 0.82, inverse vol 1.45), and model-free A2C reaches only ≈ 0.79 . No strategy breaches the CVaR limit, so the constrained and unconstrained learners are approximately equal, and the constraint costs almost nothing when there is no tail loss to prevent. The other deterministic rules trail on this split: inverse volatility (Sharpe 1.45, CVaR_{95} 0.0087, max-DD 0.041), mean–variance (0.84, 0.0198, 0.104), equal weight (0.82, 0.0163, 0.097) and risk parity (0.70, 0.0182, 0.134), while minimum variance and the min-CVaR LP share a max-DD of 0.016 and the LP has CVaR_{95} 0.0080.

7.2 Stress window: the constraint binds

Table 3 contrasts a return-greedy unconstrained learner with the CVaR-constrained allocator on the stress window (mean $\pm$ std over 5 seeds). The constraint roughly halves tail risk and drawdown while *improving* Sharpe, Sortino and Calmar, by removing the deep-loss weeks. The constrained policy rotates into the cash/rates leg under stress, consistent with the lower drawdown and breach depth below.

A paired block bootstrap of the CVaR_{99} difference (constrained minus unconstrained, 2000 resamples, block 4) gives -0.033 , with 95% bias-corrected and accelerated (BCa) interval $[-0.035, -0.022]$ and $p \approx 0$, so the tail-risk reduction is statistically significant. We

Table 4: Regime-conditional Sharpe/CVaR₉₉, pooled from the 10-fold macro-ETF walk-forward (n = weeks of that regime across folds). Best per column in bold. Constrained RL beats min-variance in high-volatility weeks, and min-variance leads calm and selloff weeks.

	Calm ($n=388$)		High-vol ($n=71$)		Selloff ($n=61$)	
Strategy	Sharpe	CVaR ₉₉	Sharpe	CVaR ₉₉	Sharpe	CVaR ₉₉
RL Constrained	1.438	0.020	2.169	0.018	-0.809	0.048
RL Unconstrained	1.282	0.028	2.194	0.023	-1.127	0.066
Min-Variance	1.770	0.009	1.668	0.007	0.035	0.016
Inverse-Vol	1.721	0.011	2.803	0.009	-1.327	0.033

Table 5: Regime-switching hybrid vs. pure strategies. Hybrid uses constrained RL in high-volatility weeks ($n=80$), min-variance otherwise ($n=440$). Bootstrap 95% CI, 50,000 resamples. High-vol regime Sharpe: constrained RL 1.415 vs. min-var 0.564.

Strategy	Sharpe	95% CI	Ann. Ret.	CVaR ₉₉
Constrained RL (pure)	0.942	[0.29, 1.68]	4.0%	0.025
Min-variance (pure)	1.209	[0.55, 1.93]	2.9%	0.012
Hybrid	1.316	[0.66, 2.03]	3.5%	0.013

report BCa rather than percentile intervals because resamples of an extreme-tail statistic often omit the few crash weeks, which biases a percentile interval toward zero. The constrained allocator also eliminates portfolio-constraint violations ($7.2 \rightarrow 0.0$ per run).

7.3 Walk-forward across regimes (10 OOS folds, $\approx 2010\text{--}2024$)

With refits every 52 weeks on a rolling 312-week window and 10 concatenated out-of-sample folds, the constraint beats unconstrained RL out of sample on every metric: Sharpe +18% ($0.70 \rightarrow 0.83$), CVaR₉₉ -20% ($0.0416 \rightarrow 0.0331$) and max-DD -25% ($15.1\% \rightarrow 11.4\%$). Inverse volatility (Sharpe 1.06, max-DD 0.070, CVaR₉₉ 0.0183) and minimum variance (1.45, 0.057, 0.0106) remain ahead in absolute terms. The fold-level Wilcoxon on CVaR₉₉ is marginal on the macro universe alone ($p = 0.06$, with 5 of 10 folds improved), and the reduction concentrates in **selloffs** (CVaR₉₉ $0.066 \rightarrow 0.048$, max-DD $0.218 \rightarrow 0.137$, Table 4). Minimum variance is therefore the strongest absolute performer, a null result on “beat the optimiser.” A two-universe 40-fold analysis raises the significance to $p = 8.8 \times 10^{-6}$ (Section 7.4).

When RL adds value: regime-conditional breakdown. High-volatility periods ($n=71$) are where the constrained policy performs best relative to minimum variance: Sharpe 2.169 versus 1.668, with CVaR₉₉ within 0.012 of minimum variance’s floor. In **calm** markets ($n=388$) and **selloffs** ($n=61$) minimum variance dominates (selloff Sharpe 0.035 vs. constrained RL -0.809): the overlay adds value when volatility is elevated but has not yet collapsed into a full drawdown, and stops adding value once liquidity deteriorates. The constraint still cuts tail risk relative to unconstrained RL in *all three* regimes (-29.1% calm, -20.7% high-vol, -27.4% selloff), which points to a structural property of the architecture rather than a regime-specific artefact.

Table 6: Walk-forward CVaR₉₉, two universes, 26-week folds. Both pass Wilcoxon and BH. Pooled $p=8.8 \times 10^{-6}$ (3.3×10^{-4} clustered by calendar). Ties, where the budget never binds: macro 11, sector 2. PSR: Probabilistic Sharpe Ratio (constrained).

universe	folds	CVaR ₉₉ ^{unc}	CVaR ₉₉ ^{con}	improved	Wilcoxon p	PSR
macro ETF (7)	20	0.0143	0.0102	9/20	0.0077	0.997
sector (10)	20	0.0336	0.0250	17/20	0.0003	0.990
pooled	40	—	—	26/40	8.8×10^{-6}	—

7.4 Statistical significance across two universes

To confirm that the effect holds beyond a single asset universe, we add ten S&P sector ETFs (XLK, XLF, ..., plus a cash leg), shorten the walk-forward test fold to 26 weeks to obtain more observations (20 folds per universe), and pool the fold-level CVaR₉₉ differences (Table 6). **Both universes individually show significant fold-level reduction:** macro-ETF Wilcoxon $p = 0.0077$ (9 folds improved, 11 tied, none worse), sector Wilcoxon $p = 0.0003$ (17 improved, 2 tied, 1 worse), and Benjamini–Hochberg [6] rejects the null on *both* (BH-reject = True). A tied fold is one where the budget never binds, so the constrained and unconstrained learners coincide. Pooling all 40 folds gives Wilcoxon $p = 8.8 \times 10^{-6}$ and a bootstrap 95% CI of $[-0.0089, -0.0041]$ on the mean fold difference, and 26 of the 27 untied folds improve (sign test $p = 4 \times 10^{-7}$). Because the two universes share a calendar, we also average their differences within each calendar fold. The resulting 20 clusters give Wilcoxon $p = 3.3 \times 10^{-4}$ (exact sign-flip $p = 3.8 \times 10^{-5}$), and moving-block bootstrap intervals over consecutive clusters exclude zero. The Probabilistic Sharpe Ratio [4] of the constrained allocator is ≥ 0.99 on both universes. The out-of-sample *Sharpe advantage over classical baselines*, however, does not survive a Deflated-Sharpe correction [5], which discounts a Sharpe ratio’s apparent significance by the number of strategy configurations searched before reporting it. The pooled-significant effect is on *tail risk*, consistent with a safety layer rather than an alpha source. Per-universe mean fold Δ CVaR₉₉ is -0.0041 (macro-ETF) and -0.0087 (sector).

7.5 Generalisation to sector ETFs

Re-running the stress study unchanged on the ten-sector universe (mean over 3 seeds) replicates the main effect. The constraint cuts CVaR₉₉ by 30% ($0.109 \rightarrow 0.076$), CVaR₉₅ by 35% ($0.058 \rightarrow 0.037$), maximum drawdown by 30% ($0.295 \rightarrow 0.208$), and the breach rate by 47% ($0.75 \rightarrow 0.40$), at *no* Sharpe cost ($0.752 \rightarrow 0.748$). A paired block bootstrap of the CVaR₉₉ difference gives -0.044, with 95% BCa interval $[-0.050, -0.032]$ and $p \approx 0$. The one difference from the macro universe is that there the constraint *improved* Sharpe (the avoided tail dominated), whereas on sectors it is Sharpe-neutral. The effect that replicates is the tail and drawdown reduction rather than a return gain.

7.6 Robustness and ablations

The CVaR₉₉ reduction holds in 7/7 perturbations, even at $3\times$ costs (CVaR₉₉ 54% lower, Sharpe 63% above unconstrained). Ablations (Table 7) reveal a **monotone pattern** (Figure 2, right): tightening the tail budget from loose ($\alpha = 0.95$, limit 1.92%) through base (1.2%) to tight ($\alpha = 0.99$ or 0.72%) *simultaneously* improves CVaR₉₉ and Sharpe at every step. The tightest configuration reaches

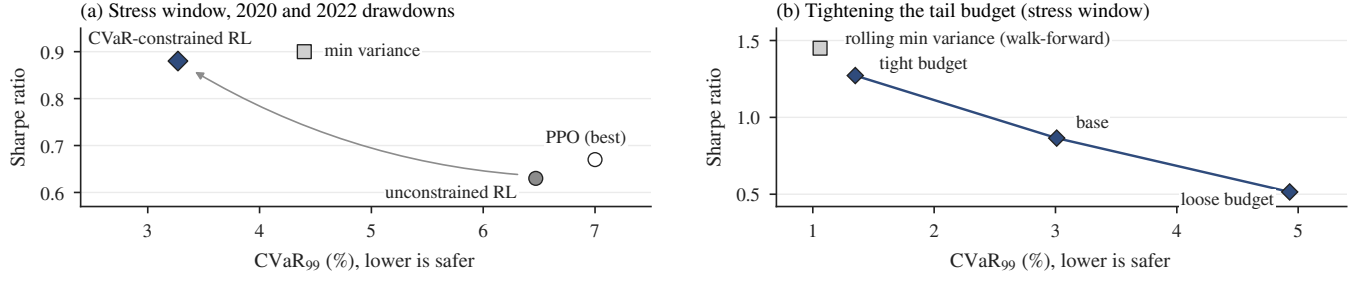


Figure 2: The CVaR constraint as a safety layer. Left. Stress-window risk–return plane (lower CVaR₉₉ is safer). The CVaR-constrained allocator (diamond) moves away from the unconstrained learner (circle). Sharpe +40%, CVaR₉₉ −49%, max-DD −43%, and portfolio-constraint violations reduced from 7.2 to zero. **Right.** Ablation monotonicity: tightening the tail budget from loose to tight improves both CVaR₉₉ and Sharpe. The tightest configuration (Sharpe 1.27, CVaR₉₉ 1.4%) is competitive with the rolling minimum-variance optimiser (square, walk-forward) without sacrificing the safety guarantee.

Table 7: Tail-budget and α ablations, macro-ETF universe, stress window. Against the base (anchor, $\alpha=0.95$, return objective), the tighter budget and $\alpha=0.99$ improve and the looser one does not.

variant	Sharpe	CVaR ₉₉	max-DD
base (anchor, $\alpha=0.95$, return obj.)	0.865	0.0301	0.111
no anchor	0.826	0.0318	0.108
$\alpha = 0.90$	0.684	0.0452	0.172
$\alpha = 0.99$	1.251	0.0146	0.053
tighter risk budget	1.272	0.0135	0.041
looser risk budget	0.515	0.0493	0.230
Sharpe objective	0.791	0.0273	0.108

Sharpe 1.27 and CVaR₉₉ = 1.4%, which is competitive with rolling min-variance (Sharpe 1.45) without relaxing the guarantee. Tighter risk limits commonly hurt return, whereas here they help, because the constraint’s gradient pressure acts only on deep-loss weeks and so removes the largest losses while barely touching the rest of the return distribution.

Features. Rolling factor betas alone slightly hurt performance. Adding macro state gives Sharpe +0.02 (3-seed mean) but worsens CVaR₉₉ (0.029 → 0.031).

Diagnostics: behaviour rather than metric gaming. Three diagnostics confirm that the constraint changes *behaviour* rather than gaming the tail metric: the unconstrained allocator concentrates and holds through the 2020/2022 drawdowns, the constrained one rotates into the cash/rates leg and holds a lower, steadier rolling CVaR estimate, and the tail control is paid for with modestly higher turnover, not leverage or a single-asset bet. The policy stays long-only, diversified, and within the turnover cap.

7.7 Tuned model-free RL remains weak

We sweep 36 PPO and 8 SAC configurations, select each by validation Sharpe, and evaluate on the stress window (Table 8). Tuning does not rescue either family. The selected PPO overfits validation (Sharpe 1.70) and then posts the *worst* OOS tail (Sharpe 0.67, CVaR₉₉ = 0.070), while SAC collapses to near-zero turnover (Sharpe 0.39). Both are *generalisation* failures: with 887 weekly observations, score-function and off-policy RL have too few effective samples.

Table 8: Best-tuned model-free RL (selected on validation Sharpe, 3-seed mean) vs. differentiable allocators and minimum variance, stress window. Best in bold.

strategy	Sharpe (↑)	CVaR ₉₉ (↓)	max-DD (↓)	turnover (↓)
minimum variance	0.90	0.044	0.113	0.011
diff. constrained	0.91	0.032	0.113	0.039
diff. unconstrained	0.60	0.062	0.212	0.023
PPO (best)	0.67	0.070	0.225	0.028
PPO (best) + CVaR	0.63	0.047	0.160	0.162
SAC (best)	0.39	0.033	0.185	0.005
SAC (best) + CVaR	0.18	0.035	0.205	0.004

The constrained allocator matches minimum variance on Sharpe (0.91 vs 0.90) while reducing CVaR₉₉ by 27% (0.032 vs 0.044).

7.8 Evaluation-protocol reversal

To probe how far the choice of *evaluation protocol* alone can move conclusions, we add a third, high-dimensional universe, **31 cross-asset, sector, and industry ETFs** (same provenance, clock, costs), where covariance estimation is harder and a learner has, a priori, more scope to add value. On a single stress split (train pre-2018, test through 2024) the constrained learner appears to *beat every optimiser*: Sharpe 0.85 versus minimum variance 0.23 (Table 9). The same minimum-variance optimiser is instead the *best* method by a wide margin under a rolling walk-forward (20 OOS folds, refit every 26 weeks): Sharpe **1.40** versus the learner’s 0.74, so the ranking *inverts*. The stress-split “win” was an artefact of one fixed period, and rolling refits remove it entirely. It is not an artefact of an unshrunk covariance on 31 assets: on the same split, Ledoit–Wolf minimum variance [16] reaches Sharpe 0.25 and the min-CVaR LP 0.26. The constraint’s tail-risk effect survives both protocols (it still cuts CVaR₉₉ versus the unconstrained learner fold-by-fold, Wilcoxon $p = 6 \times 10^{-4}$), but the apparent *beat-the-optimiser* result does not. We treat rolling walk-forward with Deflated-Sharpe [5] as the reporting standard, and flag single-stress-window comparisons of learned allocators against optimisers as unreliable.

8 Discussion

Is the learner just minimum variance under another name? The answer differs by arm. The *unfiltered* learner is far from it: its mean per-rebalance L_1 weight distance to the rolling min-variance path

Table 9: Sharpe ratios, same 31-asset universe, under two protocols.

protocol	min-var	inv-vol	constr. (ours)	unconstr.
stress split	0.23	0.38	0.85	0.70
walk-forward (OOS)	1.40	0.78	0.74	0.56

is 0.89 (of a maximum 2.0), with correlation 0.20 and a different top-weighted asset in 97% of weeks, so the resemblance that the tightest ablation suggests (Figure 2, right) lies in *aggregate risk statistics* rather than holdings. The filtered arms sit closer (L_1 0.48 to 0.61, correlation 0.72 to 0.79), buying compliance partly by occupying the low-volatility region, and from there dominate minimum variance on Sharpe and breach depth (Section 5).

For practitioners, (i) tail limits de-risk *proactively*, as the CVaR constraint shifts to cash two weeks before the COVID trough (max drawdown 11.8% vs 20.6% unconstrained), (ii) the constraint is best used as a *high-volatility overlay* (Section 4), and (iii) a near-zero multiplier signals an inert constraint, so coupling should be verified and out-of-sample folds used before an optimiser edge is claimed.

9 Scope

The pure learned allocator does not beat rolling minimum variance on Sharpe out of sample, and a Deflated-Sharpe correction removes its apparent Sharpe edge over classical baselines, so the hybrid’s +0.107 gain is regime-confined. The study covers two equity-centric universes at weekly frequency with proportional costs (no market impact or borrow), macro features are availability-lagged but not vintage, and the protocol reversal is shown on a single 31-asset window. The filter’s Gaussian CVaR estimate can understate fat-tailed losses, and the constrained learner trails minimum variance in selloff weeks (Table 4). These limitations bound the claims to a tail-risk safety layer and transferable evaluation lessons rather than a state-of-the-art allocator.

10 Conclusion and future work

Our differentiable CVaR-constrained allocator reliably reduces tail risk (26 of 27 untied folds, $p = 8.8 \times 10^{-6}$), eliminates portfolio-constraint violations in stress and de-risks *before* crisis troughs, and a regime-switching hybrid reaches Sharpe 1.316 vs 1.209 for min-variance (overlapping intervals). Two failure modes transfer to any constrained financial RL study, a **Lagrangian constraint-coupling failure** and an **evaluation-protocol reversal**, and a decision-time CVaR filter delivers the out-of-sample compliance that neither rescaling the dual nor recalibrating the budget could. Constrained RL is best positioned as a *safety layer* and *high-volatility overlay*: the effects that survive are tail-risk control and budget compliance, not alpha.

Acknowledgments

All figures and tables are reproducible from the open pipeline (leak-free checks, 81 unit tests, fixed seeds, config snapshots, pinned dependencies) at https://github.com/nl2992/ICAIF_cvar-rl-portfolio-allocator.

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